THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** synapse	Chemical transmission
2	**(a)** cerebellum	Hind-brain
3	**(c)** cytokinin	High in fruits and seeds
4	**(a)** Correct	Reflex arc
5	**(b)** Student A is correct	Pancreas
6	**(b)** thyroxine	Goitre if deficient
7	**(c)** chemotropism	Chemical stimulus
8	**(c)** adrenaline	Adrenal gland
9	**(d)** change in water content of cells	Not a growth movement
10	**(a)** cranium (skull) and cerebrospinal fluid	Fluid-filled balloon

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The pathway followed by a nerve impulse in a reflex action: receptor → sensory neuron → spinal cord → motor neuron → effector.	2
12	Auxin, gibberellin (cytokinin)	1+1
13	Iodine is needed for thyroxine; deficiency causes goitre.	2
14	False — hormones are secreted into the blood and act at distant target sites.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dendrites: receive impulse; cell body: contains nucleus; axon: carries impulse away; nerve ending: releases chemicals at the synapse	3
16	Auxin synthesised at the shoot tip diffuses to the shaded side; cells there elongate more, so the shoot bends towards light.	3
17	(i) pituitary — growth (ii) pancreas — regulates blood sugar (iii) testis — male secondary sexual characters	1+1+1
	OR: Nervous: electrical impulses, fast, short-lived, via nerves; hormonal: chemicals, slow, long-lasting, via blood	3
18	(i) involuntary actions — BP, salivation, vomiting (ii) thinking, memory, voluntary actions (iii) controls hunger, releases hormones to pituitary	1+1+1
	OR: By feedback mechanisms; e.g. rise in blood sugar is detected by the pancreas which releases more insulin; as sugar falls, insulin secretion decreases.	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) cerebrum, mid-brain, pons, medulla, cerebellum, spinal cord, pituitary (ii) fore-brain: thinking, sensory/motor areas; mid-brain: reflex movements of head/eyes; hind-brain: cerebellum (balance), medulla (involuntary)	2+3
	OR: (i) sudden automatic response to stimulus (ii) touching a hot object: receptor (skin) → sensory neuron → spinal cord (relay neuron) → motor neuron → muscle (hand withdrawn) (iii) quick protective response	1+3+1
20	(i) pituitary — growth hormone (dwarfism) (ii) diabetes (iii) insulin — pancreas (iv) insulin injections and controlled diet	2+1+1+1
	OR: (i) directional growth movements in response to stimulus (ii) shoot bends towards light (+ve phototropism), roots grow downward (+ve geotropism) (iii) roots grow towards water	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 6: **Control and Coordination** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** an electrical impulse	Nerve impulse
2	**(b)** medulla	Hind-brain
3	**(a)** abscisic acid	Stress hormone
4	**(b)** Incorrect — roots are positively geotropic	Grow downward
5	**(b)** Student A is correct	Thyroid is in the neck
6	**(d)** testes	Male hormone
7	**(b)** thigmotropism	Touch
8	**(a)** pituitary	Controls others
9	**(b)** vertebral column	Backbone
10	**(b)** changes in special proteins in response to impulses	Shape and arrangement change

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gap between nerve endings of two neurons; chemicals released at the axon end diffuse across and start an impulse in the next neuron.	2
12	(i) testosterone (ii) oestrogen	1+1
13	Promote growth of the stem (stem elongation), seed germination.	2
14	False — it is due to rapid change in water content of cells, not growth.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Through hormones secreted by endocrine glands directly into blood; e.g. adrenaline raises heart rate; insulin lowers blood sugar	1+2
16	Electrical impulse at the axon end releases chemicals that diffuse across the gap to the dendrite of the next neuron; chemicals are released only at the axon end	2+1
17	Secreted directly into blood; heart beats faster supplying more O_2; blood diverted to skeletal muscles; breathing rate increases; body prepared to fight or flee	3
	OR: (i) cell elongation (ii) cell division (iii) inhibits growth, wilting of leaves	1+1+1
18	Germinating seedlings in a box with a slit allowing light: shoot bends towards light; a potted plant kept horizontal: roots bend downward, shoot bends up	3
	OR: It receives information from all receptors, processes it and sends instructions; fore-brain, mid-brain, hind-brain	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) pituitary — growth hormone; thyroid — thyroxine; pancreas — insulin; adrenal — adrenaline; testis — testosterone; ovary — oestrogen (ii) swollen neck due to thyroxine deficiency; using iodised salt	3+2
	OR: (i) central (brain, spinal cord) and peripheral (cranial and spinal nerves) (ii) reflex actions; conduction of impulses to/from brain (iii) communication between CNS and body parts	2+2+1
20	(i) phototropism (ii) auxin (iii) auxin diffuses to the shaded side, cells there elongate faster, shoot bends towards light (iv) the shoot bends again towards light	1+1+2+1
	OR: (i) plants: no nervous system, slow, through hormones and water changes; animals: nervous and endocrine systems, fast, muscle movement (ii) growth-dependent — tropisms (pea tendrils); independent — touch-me-not leaves	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

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